

Surplus Production Models for Adaptive Fisher Behavior

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Introduction

A surplus production model is a simple population dynamics model used for fisheries stock assessment. The objective of a surplus production model is to estimate the Maximum Sustainable Yield (MSY) using limited data, to identify the optimum harvest effort that avoids depleting the stock. The equation for the surplus production model is:

$$B_{t+1} = B_t + rB_t(1 - B_t/K) - qEB_t$$

Where B is the stock biomass at time t r is the intrinsic growth rate K is the environmental carrying capacity (units/cell) q is the catchability coefficient E is fishing effort

The catchability coefficient (q) relates fishing effort to fishing mortality and is assumed to be constant in this equation.

However, the true value of the catchability coefficient changes with increased fishing efficiency, fish density (hyperstability or hyperdepletion), environmental changes, or technological improvements. This has the potential to impact the determined MSY of a stock, having implications for policy development and fisher economic gain. As such, my project aims to evaluate whether assuming a constant catchability coefficient (q) in a Bayesian stock assessment leads to biased estimates when fishers are adaptively targeting aggregations.

The POSEIDON model is an interactive agent-based modeling platform that serves as an interactive tool that encodes certain parameters within a dynamic spatial-visual platform. Users of the interface apply values to these parameters to evaluate the effect of various policy interventions on ecological and fisheries outcomes. Using this model, I will be able to simulate fisheries data where fishers are highly exploratory versus where fishers use explore-exploit-imitate to aggregate effort on high-density patches through patterns of exploitation and imitation.

With these two datasets, I will run six models – fitting Bayesian surplus production models with both uninformative and informative priors (assuming good biological data for the simulated fish species) to evaluate:

- 1) How does adaptive fisher behavior bias posterior estimates of MSY in a Bayesian surplus production model that assumes constant catchability?
- 2) How does this misspecification affect posterior uncertainty, including CI width?
- 3) To what extent can informative priors mitigate bias arising from misspecified catchability?
- 4) Do Bayesian models that allow for flexible or time-varying catchability better recover true MSY under adaptive fishing behavior?

Methods

The POSEIDON model was used to simulate fishing scenerios with two different types of fisher behavior. These fisher simulation runs differ only in fisher targeting behavior, all else being held constant.

In Scenerio A - fishers move randomly, catch is roughly proportional to abundance. This means that CPUE is a good index of biomass and constant q is approximately valid.

In Scenerio B - fishers use POSEIDON's adaptive explore-exploit-imitate algorithm to seek out high-density patches. Agents learn to efficiently target remaining fish aggregations, depleting nearby areas first and expanding outward. CPUE stays high as biomass drops (hyperstability), meaning the constant q assumption is violated.

The constant parameters of both simulations are:

$K = 3000$ (max fish per cell) * (2500 cells_sq * 0.85 habitable) $K = 6375000$ fish in the system

$r = 0.35$

$q = 0.002$

$MSY = rK/4$

These parameters result in a $MSY_{true} = 557,812$

Fleet size = 100

Speed = 5 km/h

Maximum of 7 days at sea

12 hours between each departure

Fuel tank size = 100,000 liters

Liters/km = 10

Gas price/liter = 0.01

Market price of fish = \$10

Hold size = 200

Each fisher has two information sharing 'friendships'

Scenerio A - Fixed behavioral probability

Exploration probability = 0.95 (high)

Imitation probability = 0.0 (no imitation)

Scenerio B - Adaptive behavioral probability

Exploration probability = 0.05 (low random exploration) Min = 0.01

Imitation probability = 1.0 (high social imitation) Increment multiplier= 0.03 (learning speed)

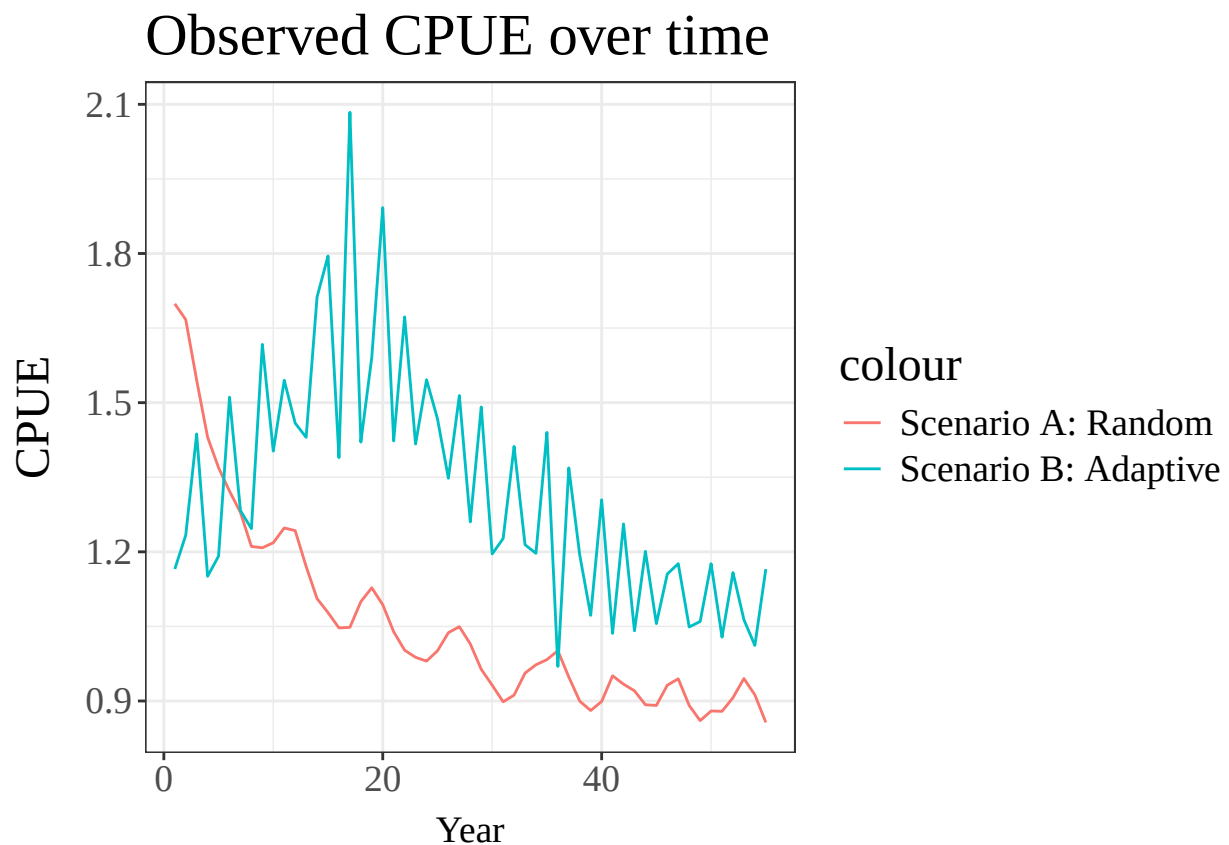
Dataset

A simulation of 60 years was run for each scenario with the given parameters. The first 5 years were removed to drop potential trends arising from initialization and to remove the learning artifact in the Scenario B, retaining years 6-60 which were then re-indexed as 1-55.

Species biomass (kg), total catch (kg), average distance from port (km), average last trip hourly income (\$), hours out, and effort were collected from the simulation. CPUE was calculated from total catch (kg) and effort. An ‘observed’ CPUE was calculated by adding adding lognormal (multiplicative) error to create realistic fisheries-style observation error.

Results

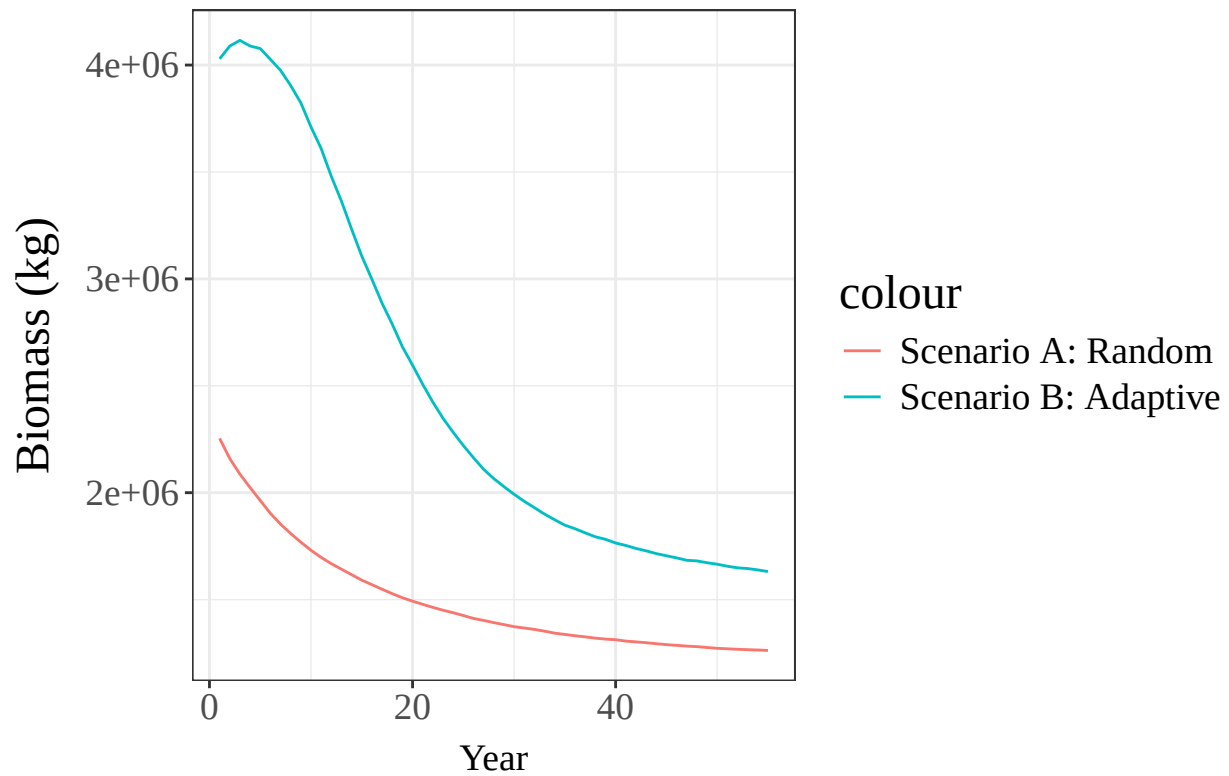
Preliminary Data Evaluation



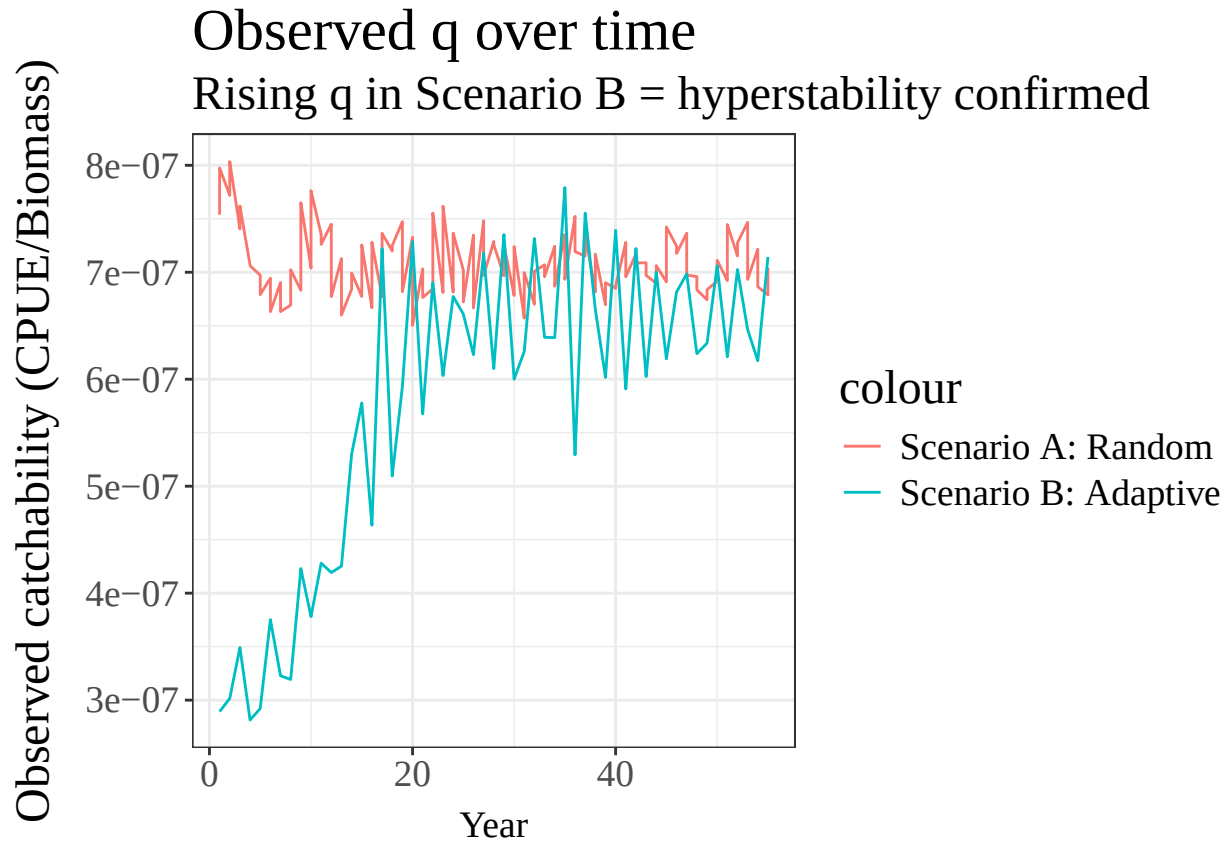
Scenario A shows a steady decrease in CPUE over time. As a result of highly exploratory behavior, the decrease in biomass (seen below) is directly tied to a decrease in CPUE.

Scenario B shows an initial increase in CPUE given higher biomass and then declines more slowly than in Scenario A. Adaptive behavior enables fishers to imitate and exploit centers of high biomass, leading to increased instability in CPUE as fishers find and deplete these centers.

Observed Biomass over time



Both Scenarios at year 0 have the same initial biomass. In Scenario A, biomass rapidly decreases initially and steadies to a point of limited decrease in later years, but remains above zero. In Scenario B, the initial biomass spike is caused by the inefficiency associated with learning. However, Scenario B shows a similar pattern of rapid decline followed by stabilization.



Scenario A represents consistency with the assumption for a standard population surplus model. The relationship between CPUE and biomass creates a largely flat trend where CPUE decline is proportional with biomass decline over time. The trend indicates that a constant q assumption is largely valid.

In contrast, Scenario B shows hyperstability. The relationship between CPUE and biomass is initially low due to learning restrictions, then increases despite decreasing biomass as a result of advantageous exploitation and imitation behavior. Spatial targeting results in a more volatile pattern as established previously. The trend indicates that a constant q assumption is not valid.

Model 1: Standard Surplus Production

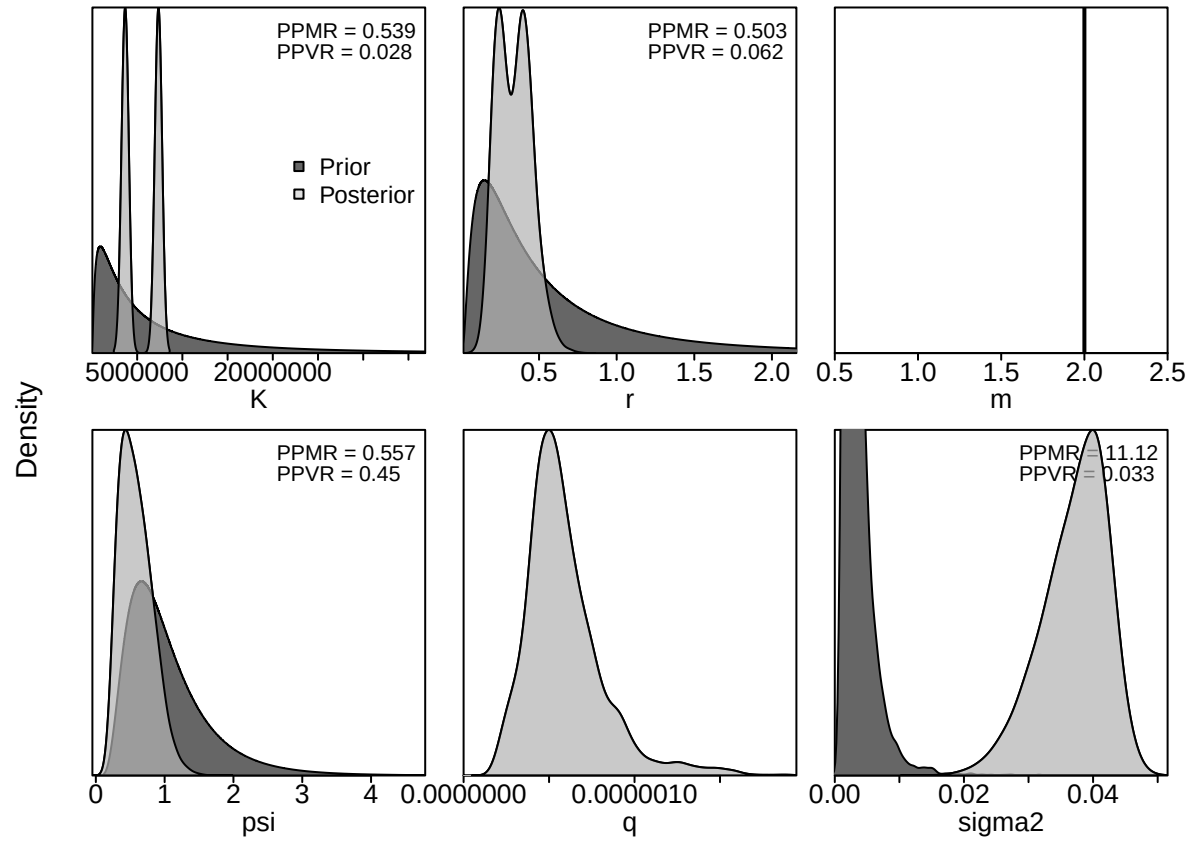
Uninformative priors were set using weak information about the system dynamics. These models encode priors that are biologically reasonable for the system but have wide standard deviation to allow for a large amount of variation from the set mean.

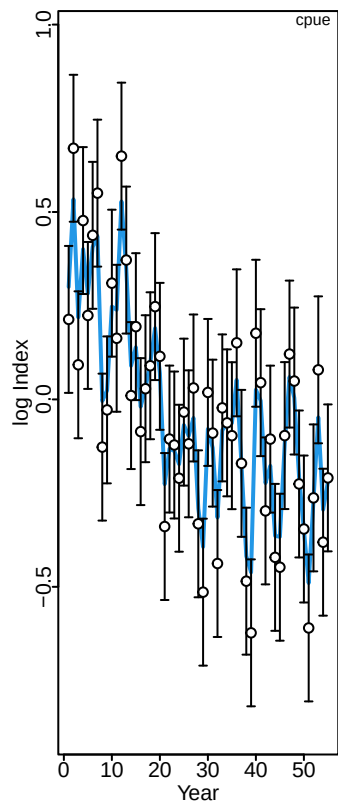
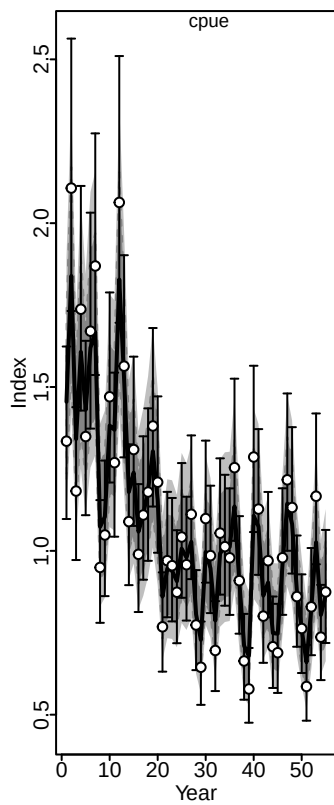
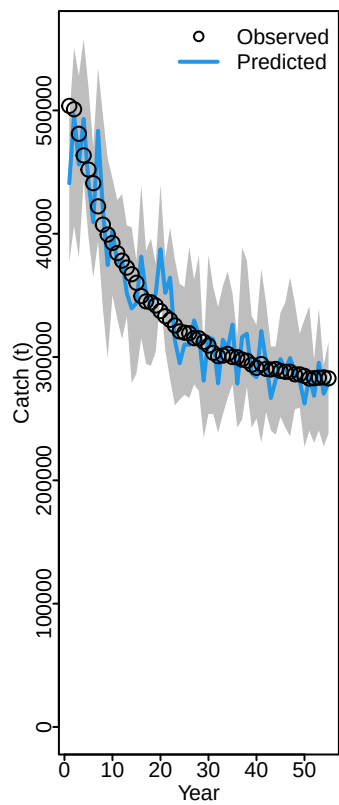
A prior of 0.4 with a standard deviation of 1 was used for r . 0.4 is considered a reasonable estimation for the growth rate of a highly productive species, with true values falling between 0 and 1. A prior of 4600000 was used for K as an estimation based on 8x the maximum catch, which tends to provide a reasonable estimate. A standard deviation of 2 allows for variance from this set mean.

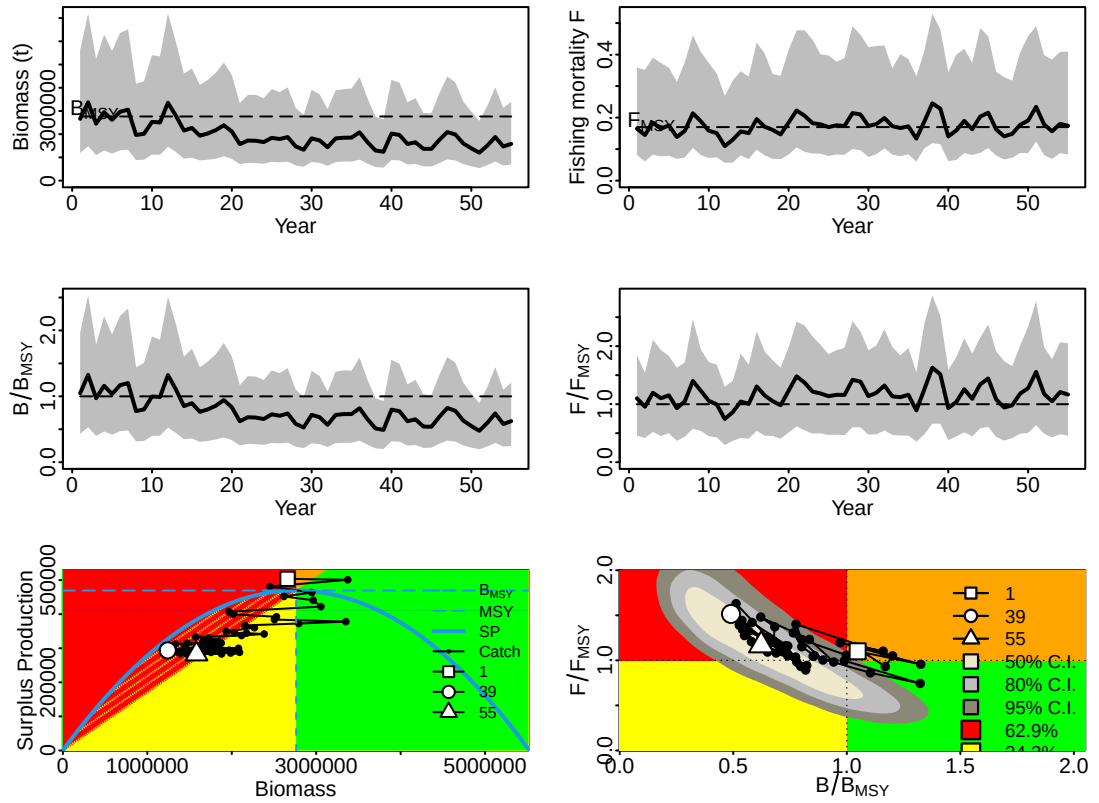
Scenario A

Application of the uninformative priors model to Scenario A (true constant q) using a JABBA Schaefer model

The posteior estimate shows a bimodal K which remains, even with increasing the number of iterations. The Kobe plot shows relative certainty that the true population is overfished and experiancing overfishing, in line with the decreasing CPUE and low biomass.



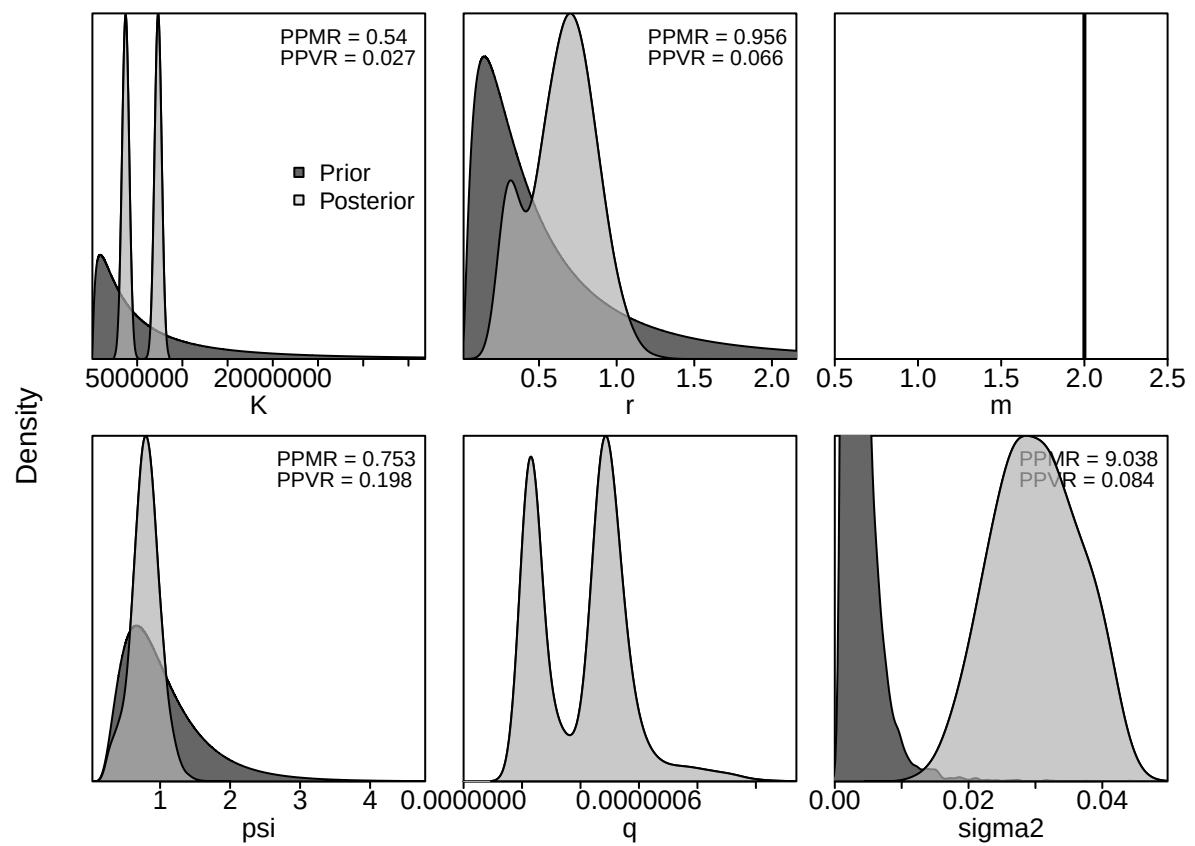


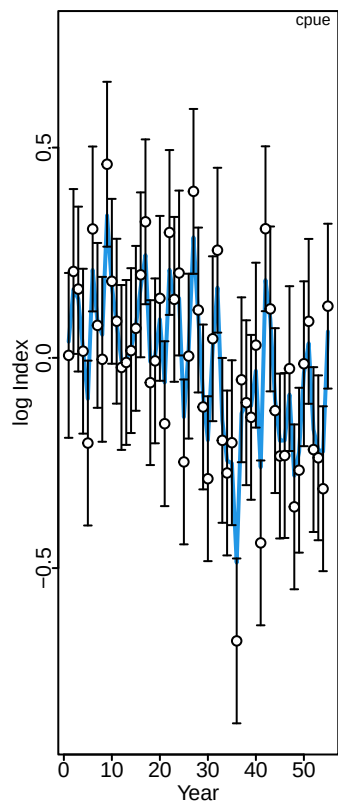
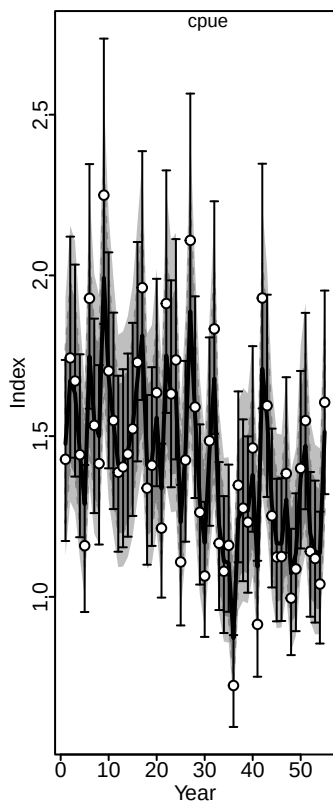
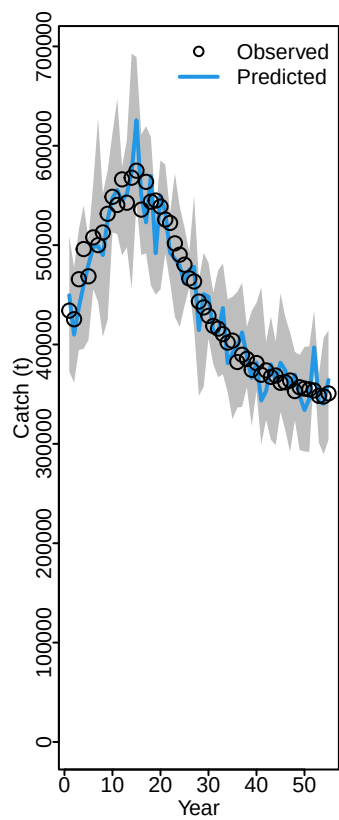


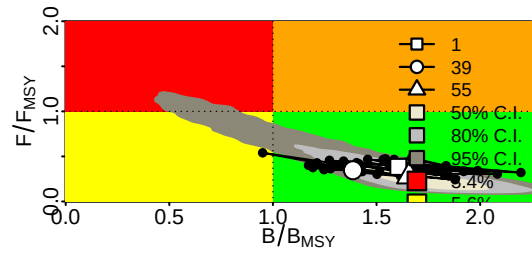
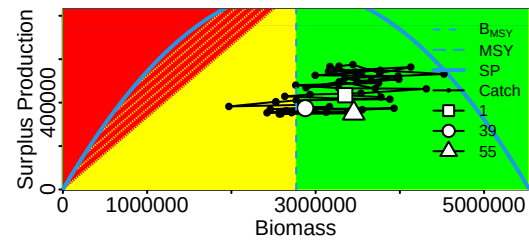
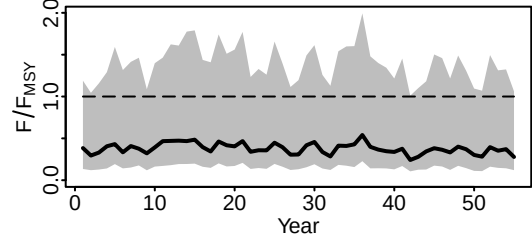
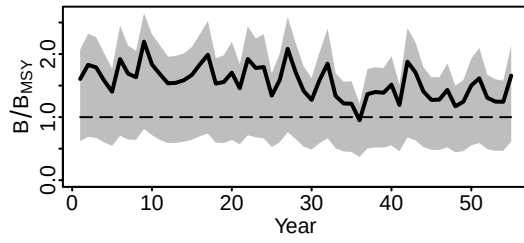
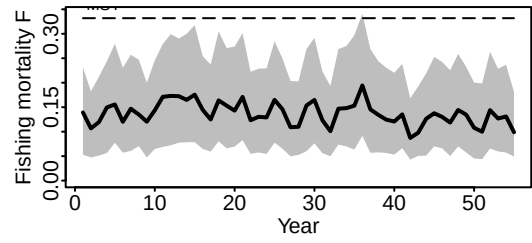
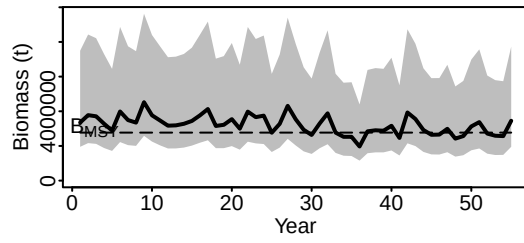
Scenario B

Application of the uninformative priors model to Scenario B (adaptive q) using a JABBA Schaefer model

Priors show a bimodal K which remains, even with increasing the number of iterations. The Kobe plot shows a high level of uncertainty, emphasizing that uninformative priors may lack strength to form an accurate estimate of stock status.

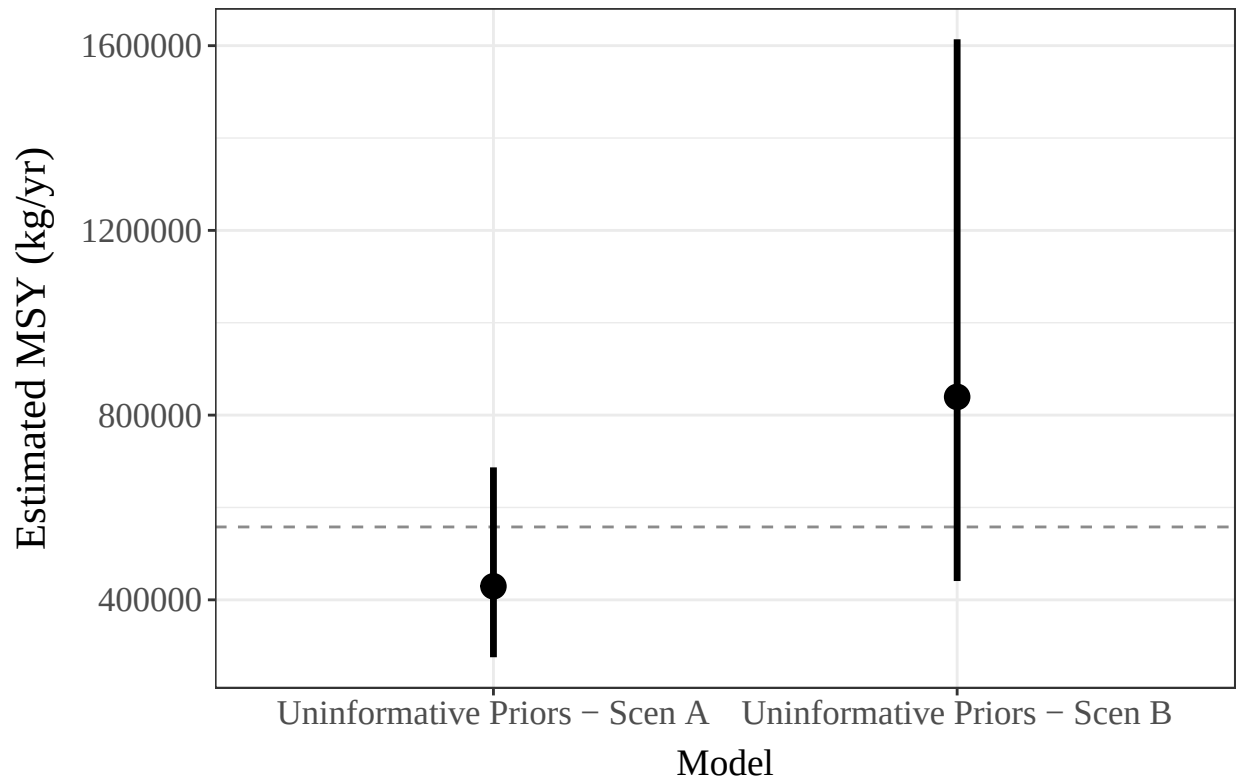






Model Comparison

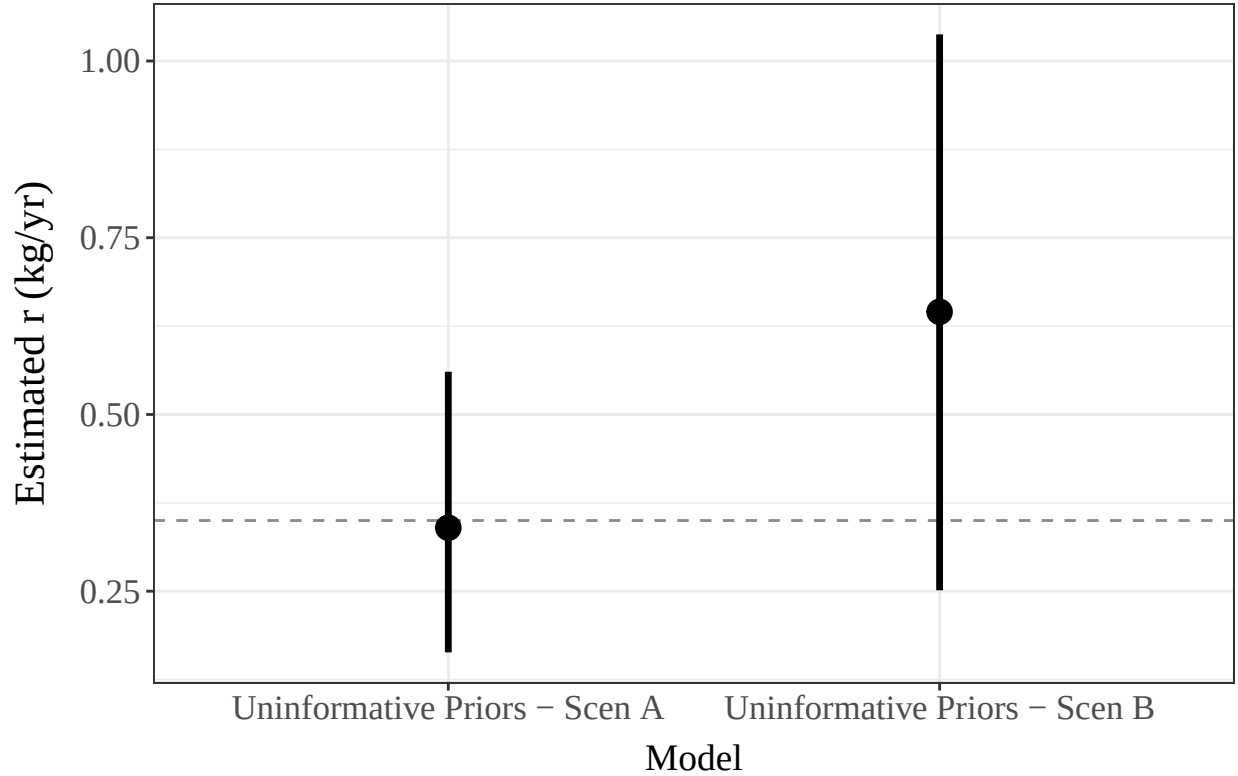
Estimated MSY compared to True MSY



Estimated K compared to True K



Estimated r compared to True r



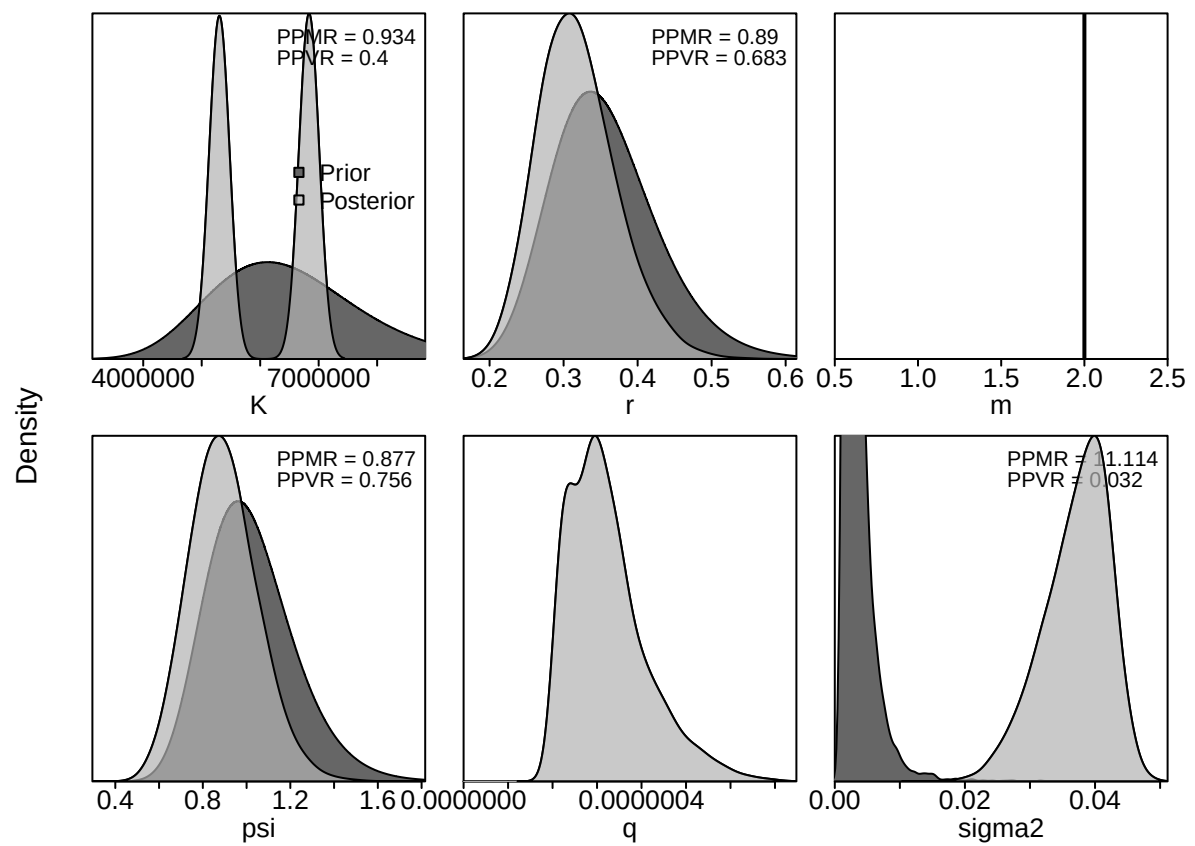
Uninformative priors do not produce highly accurate posterior coefficient estimates in either Scen A or Scen B. Both scenarios contain the true MSY within the CI. However, particularly in ScenB, the CI is very broad, as the priors are too broad to constrain uncertainty. Overarchingly, the posterior estimates are more accurate in Scen A than in Scen B, however, given the means, both models risk overharvest with estimated MSY values that are greater than true MSY.

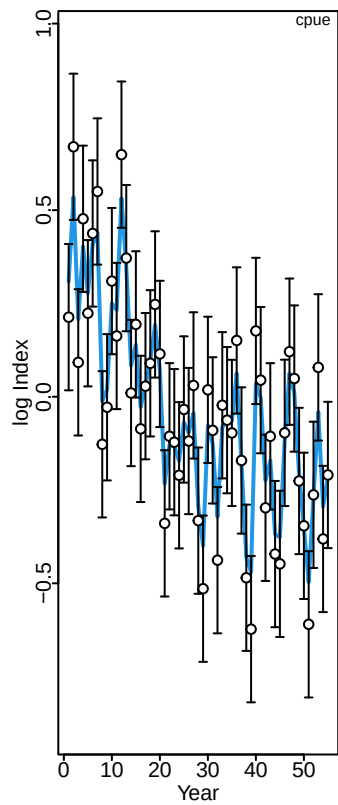
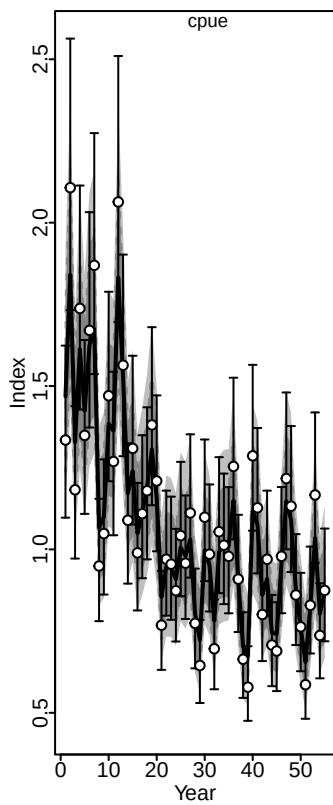
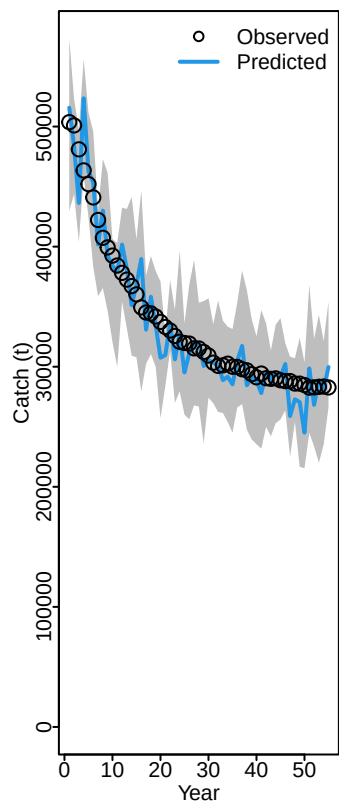
Model 2: Standard Surplus Production + Informative Priors

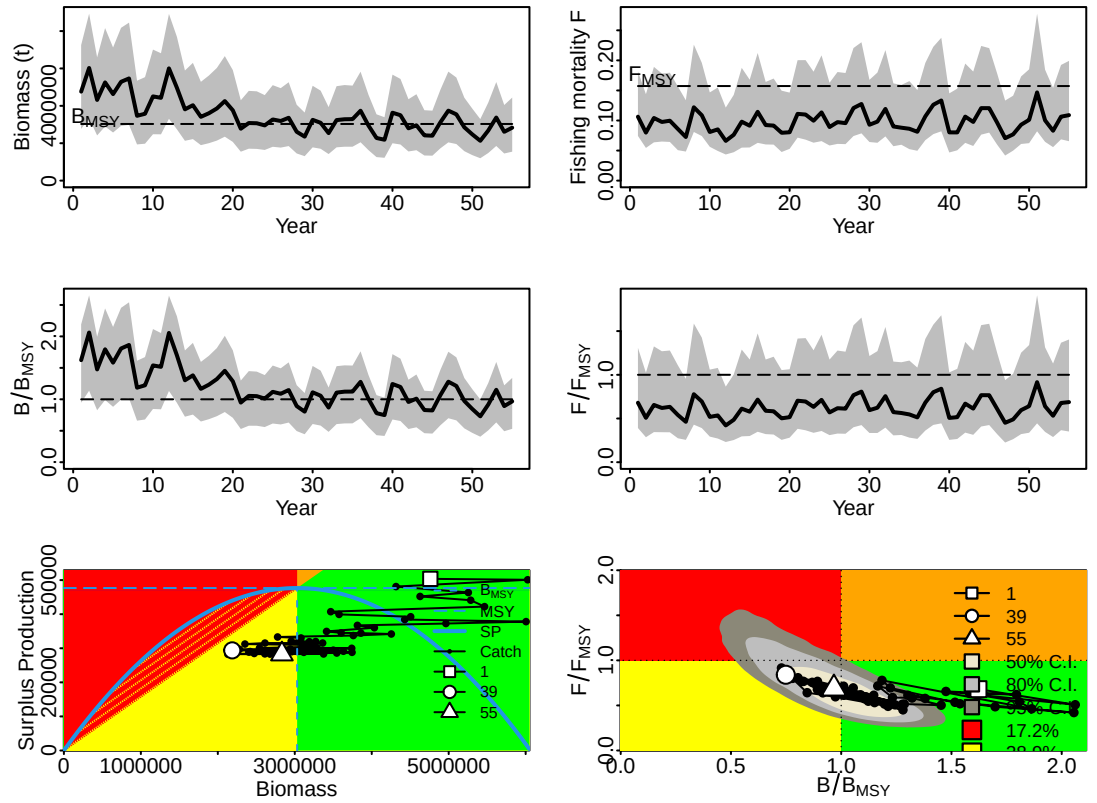
This scenario imitates a fishery in which there is greater biological certainty regarding the true r and K values. In this, the values of r and K are centered at their true value in the simulation, with reduced variation given in the standard deviation.

Scenario A

Prior fit for r and K is improved, though the posterior for K remains bimodal. Kobe plots show overfished stock, with potential for overfishing within the CI.

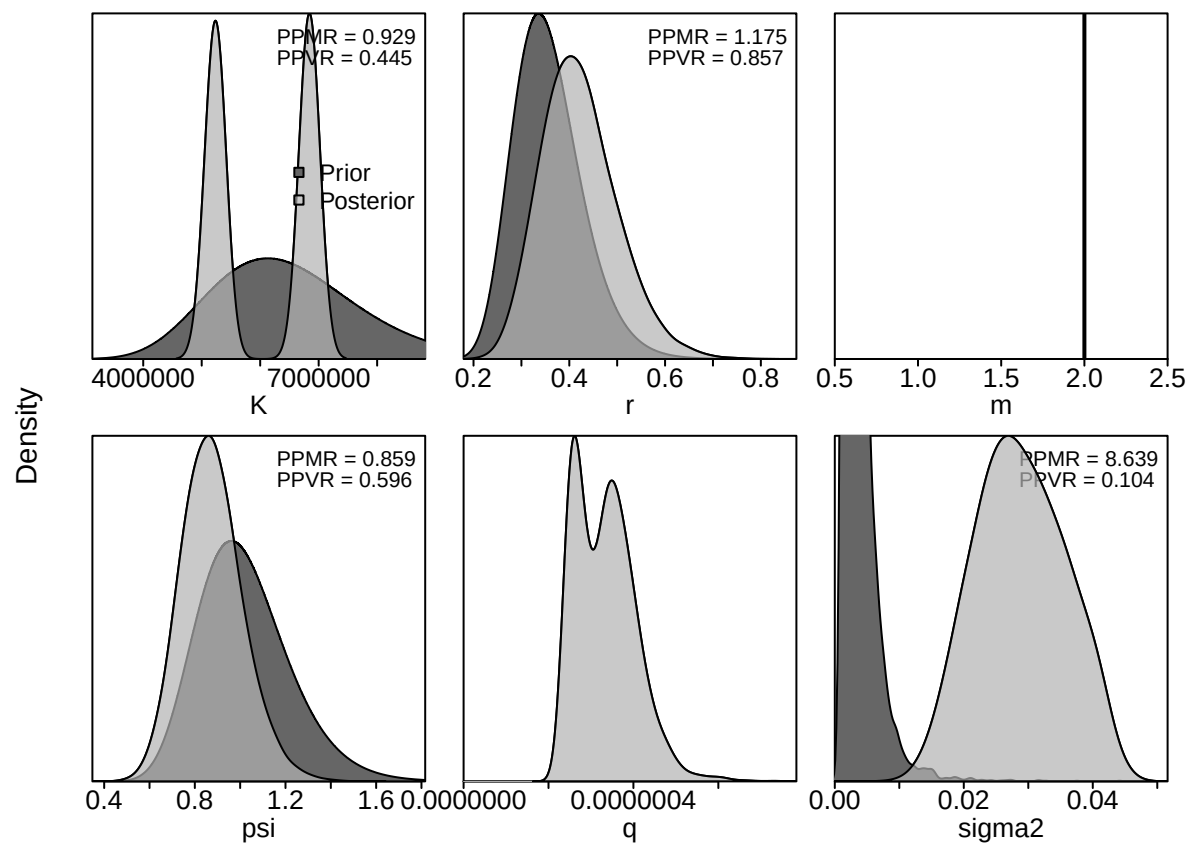


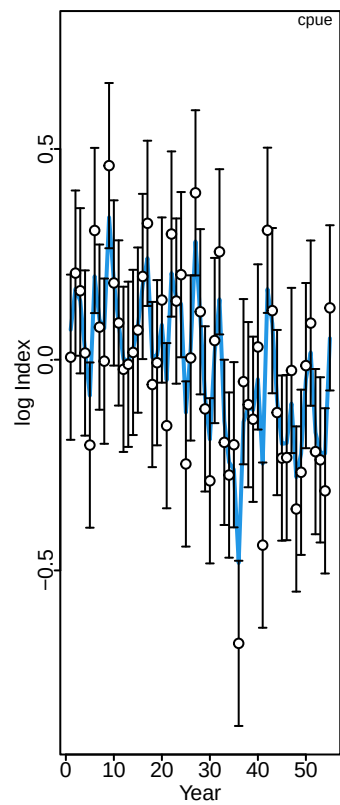
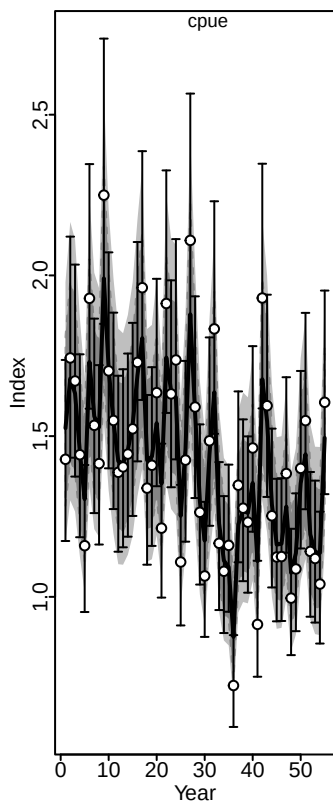
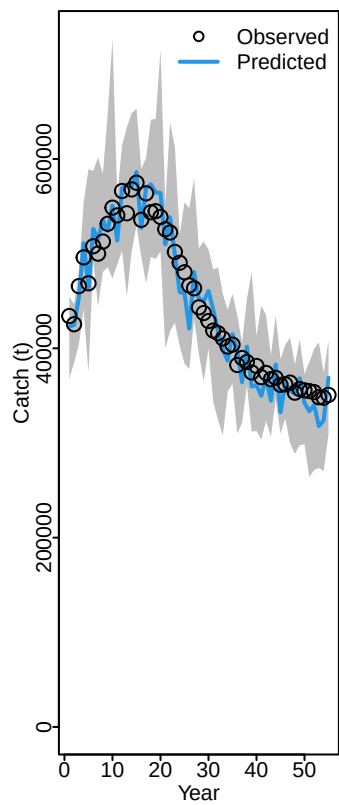


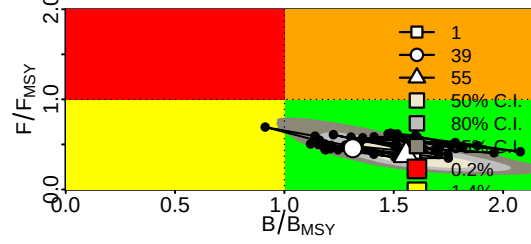
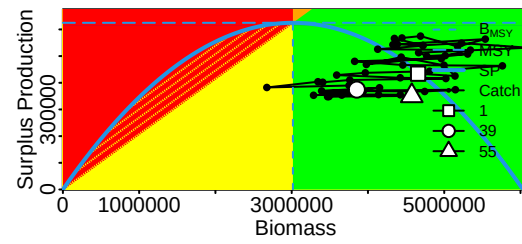
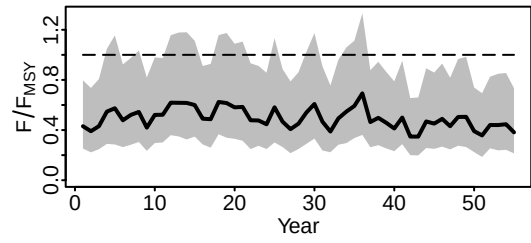
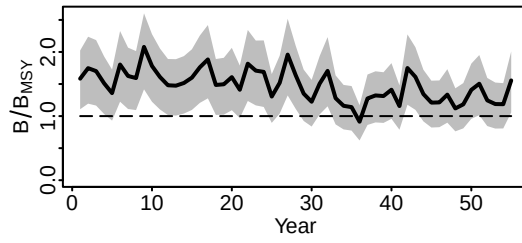
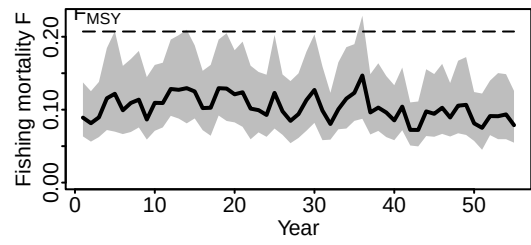
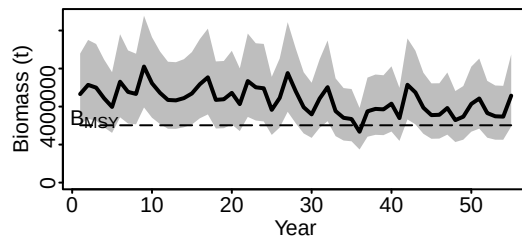


Scenario B

Prior fit for r and K is improved, though the posterior for K remains bimodal. Kobe plots show a primarily not overfished nor overfishing stock, with a more reduced CI.

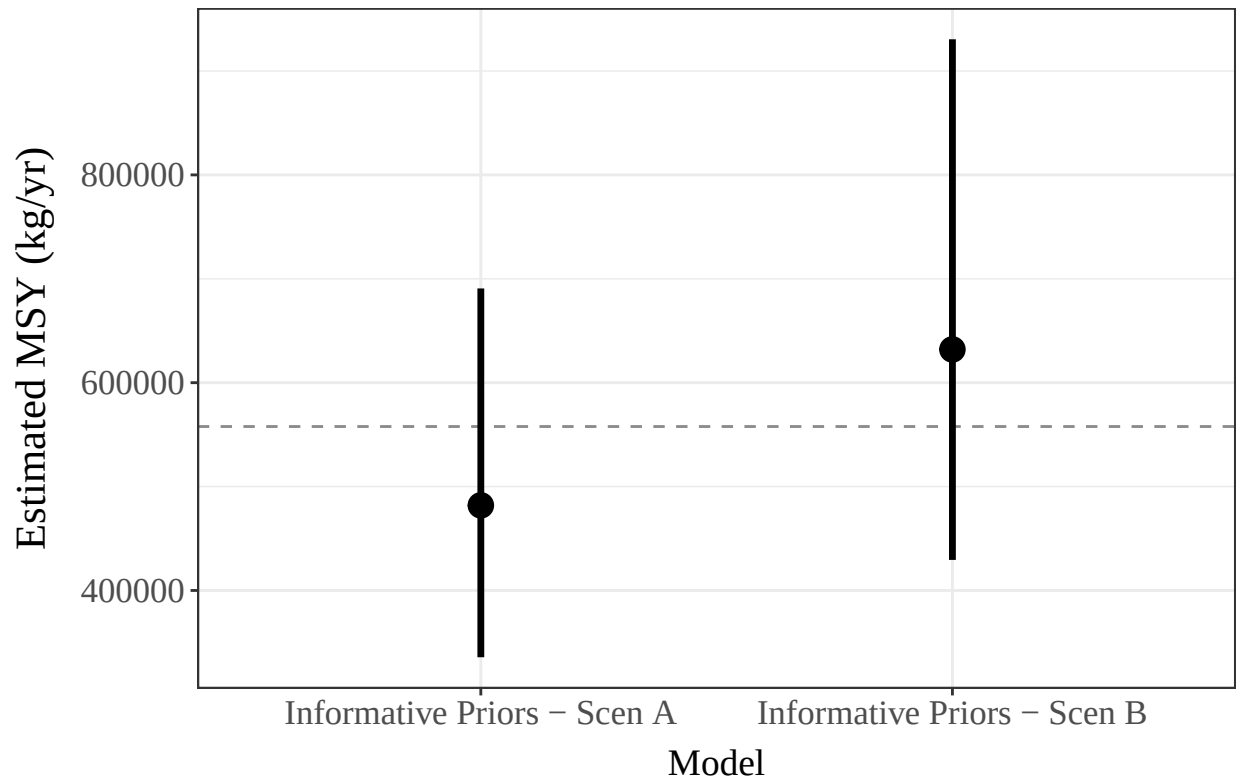




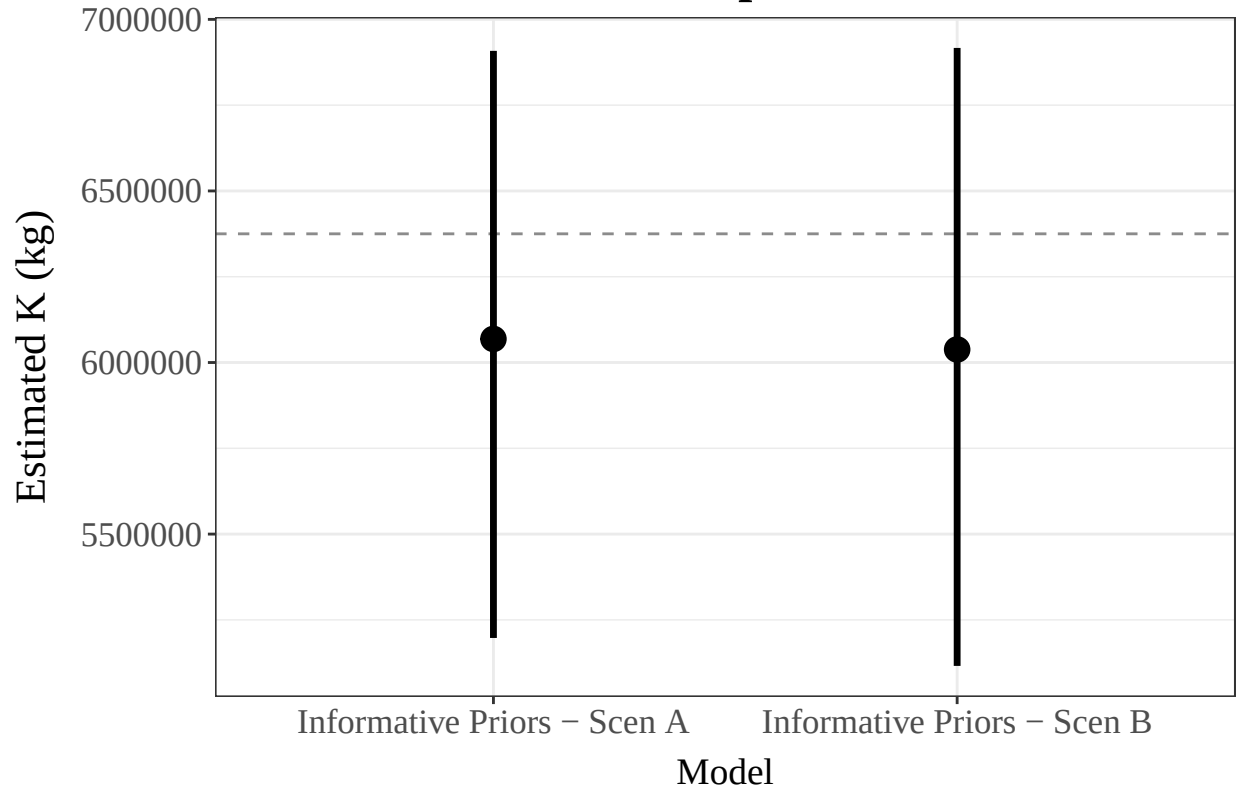


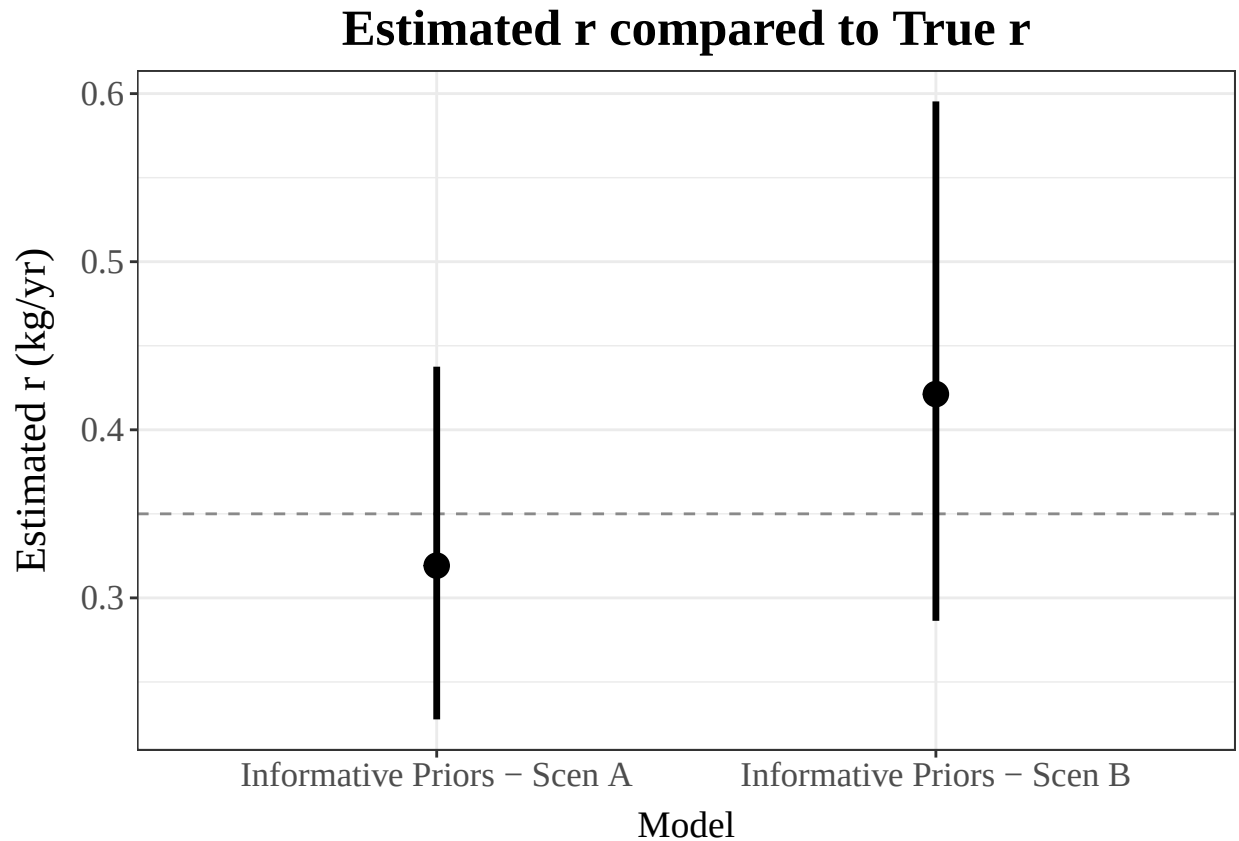
Model Comparison

Estimated MSY compared to True MSY



Estimated K compared to True K



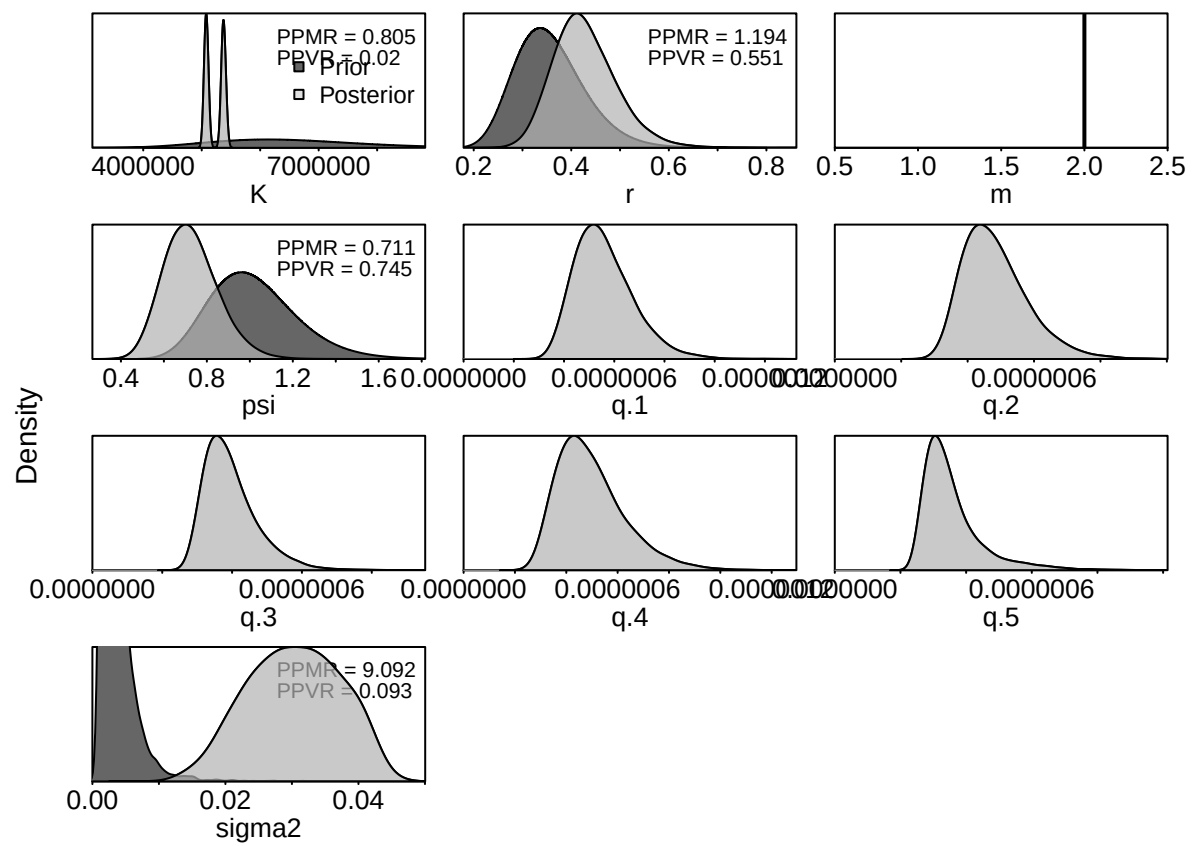


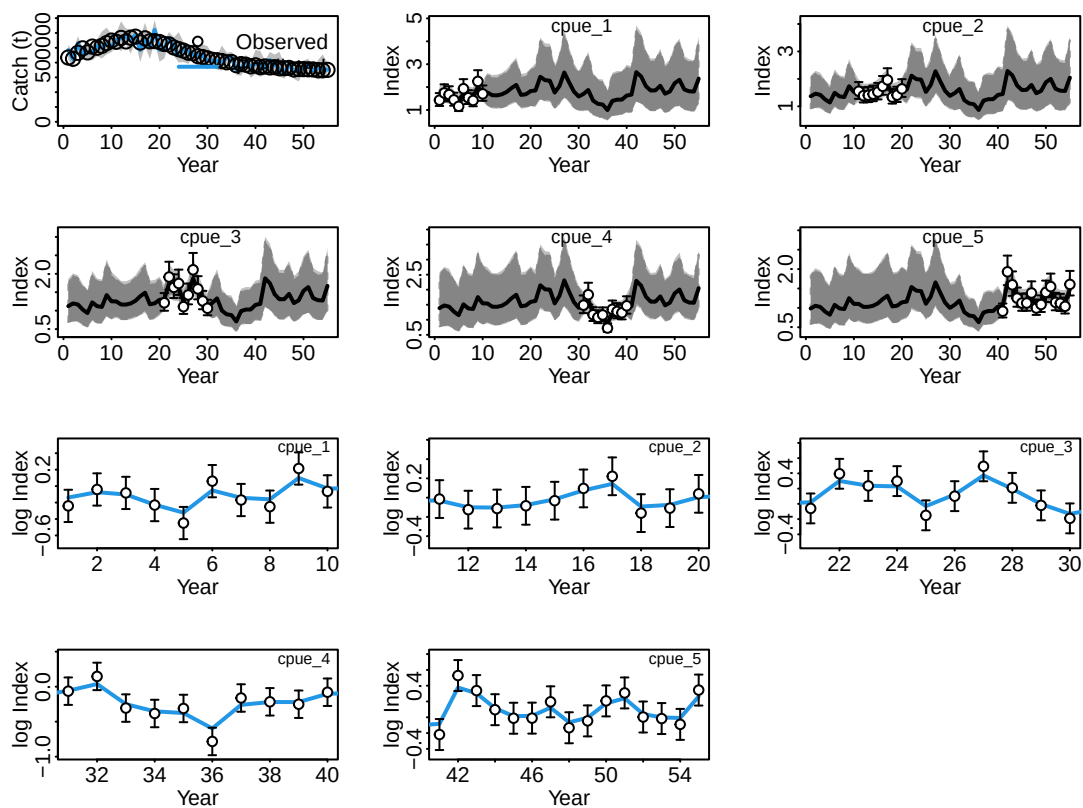
Both models show posterior estimates for MSY, K, and r that contain the true value. The mean of each of these values is closer to the true value in Scen A than in Scen B, which is to be expected. Additionally, the addition of informative priors reduces the uncertainty in the posterior, as seen in the reduced width of the confidence interval. Both models risk some extent of overharvest, as their estimate of MSY is greater than true MSY.

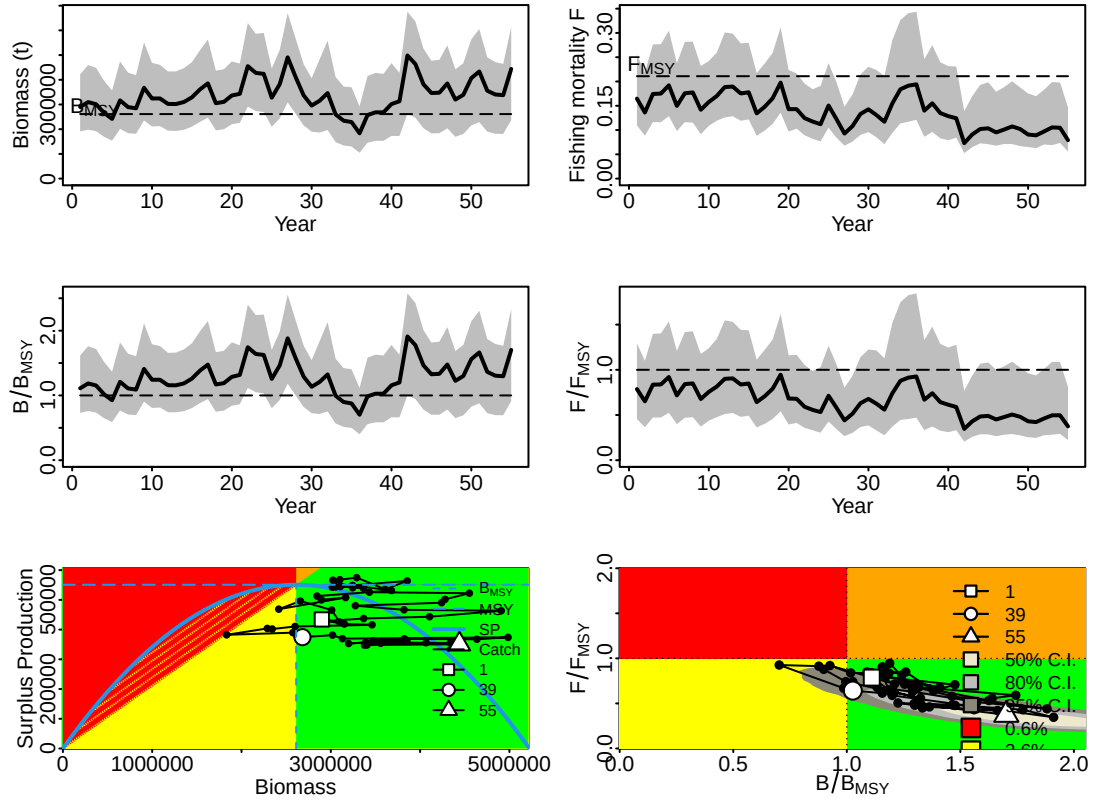
Model 3: Flexible Catchability

This model uses the `sets.q` function within JABBA to assign catchability to different CPUE indices. This subsets catchability into five different time periods.

The Kobe plot shows very broad distribution between quadrants, emphasizing uncertainty. Using a piecewise block on q limits the ability of the model to create overarching trends.







Model 4:

This model is a Stan model that encodes for a random walk on q in order to allow the model to adapt to variable fisher efficiency, rather than forcing it to be constant or piecewise.

In this, Stan is used to produce two models - a random walk and a constant q model. These are both run on the Scen B data. This allows direct comparison between the two models to whether using a random walk on q improves the fit and predictive ability of the model.

Random Walk

```
## Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
## resized.
```

variable	mean	median	sd	mad	q5	q95
r	0.4690571	0.4616973	0.0977804	0.0955560	0.3225707	0.6416
K	4325807.6437000	4232359.2000000	842501.2491010	786891.7291200	3130755.5850000	5820554.5300
sigma_q	0.7401194	0.7375866	0.0974500	0.1003831	0.5871377	0.9068
sigma_proc	0.1526610	0.1346912	0.1101100	0.1197768	0.0229379	0.3648

sigma_obs

1.5229574

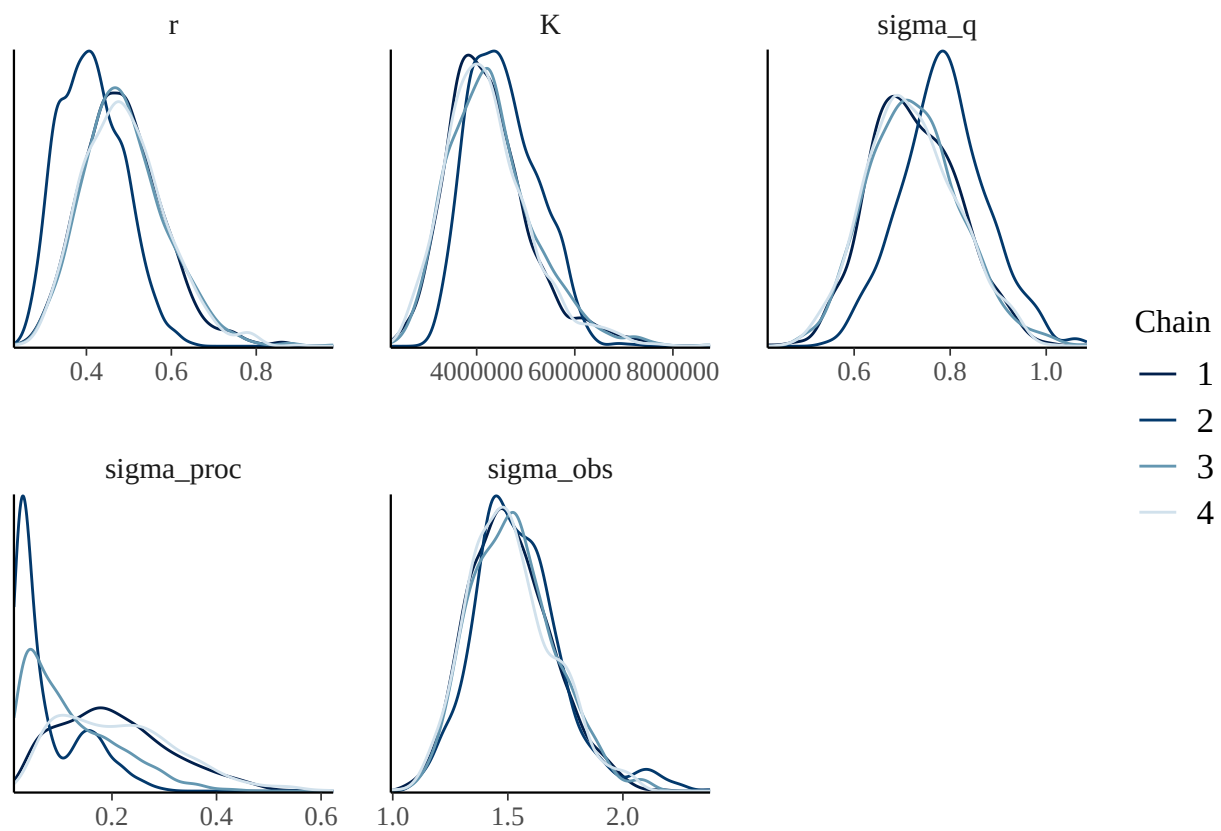
1.5056713

0.1861652

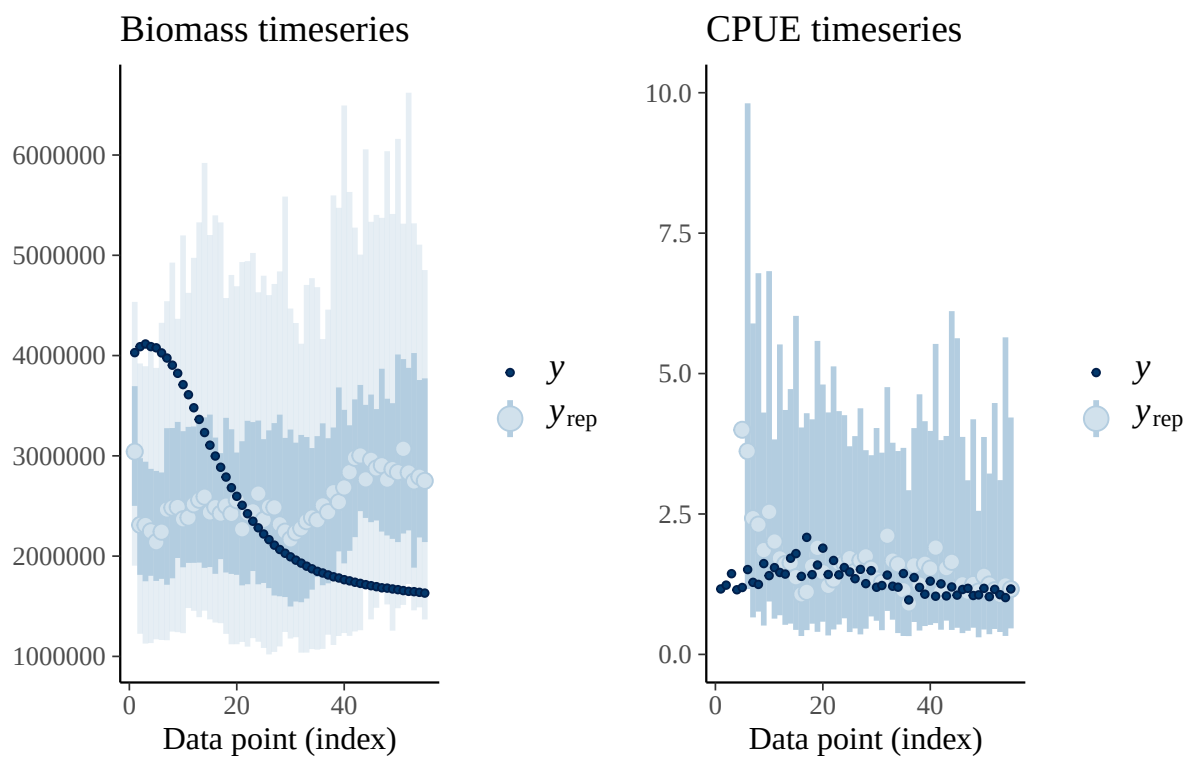
0.1804682

1.2462356

1.8552

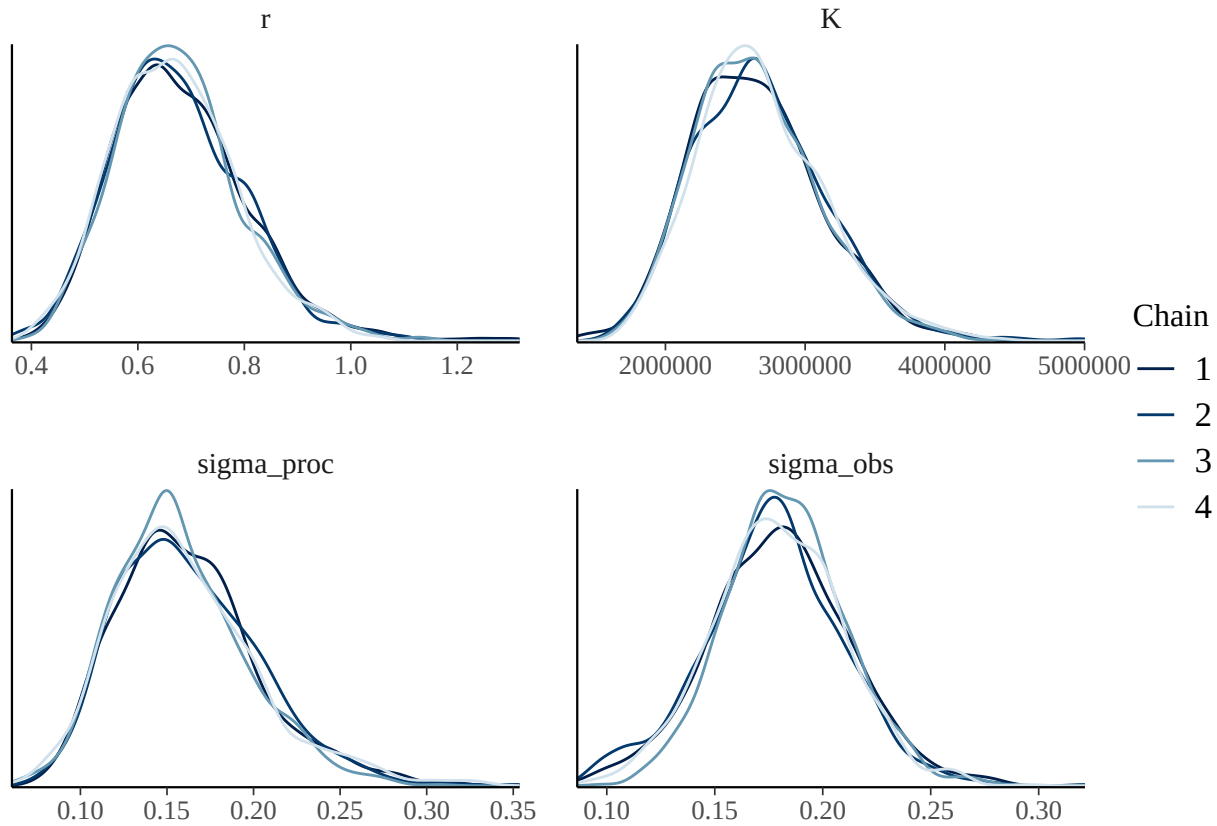


PPC: Random Walk q

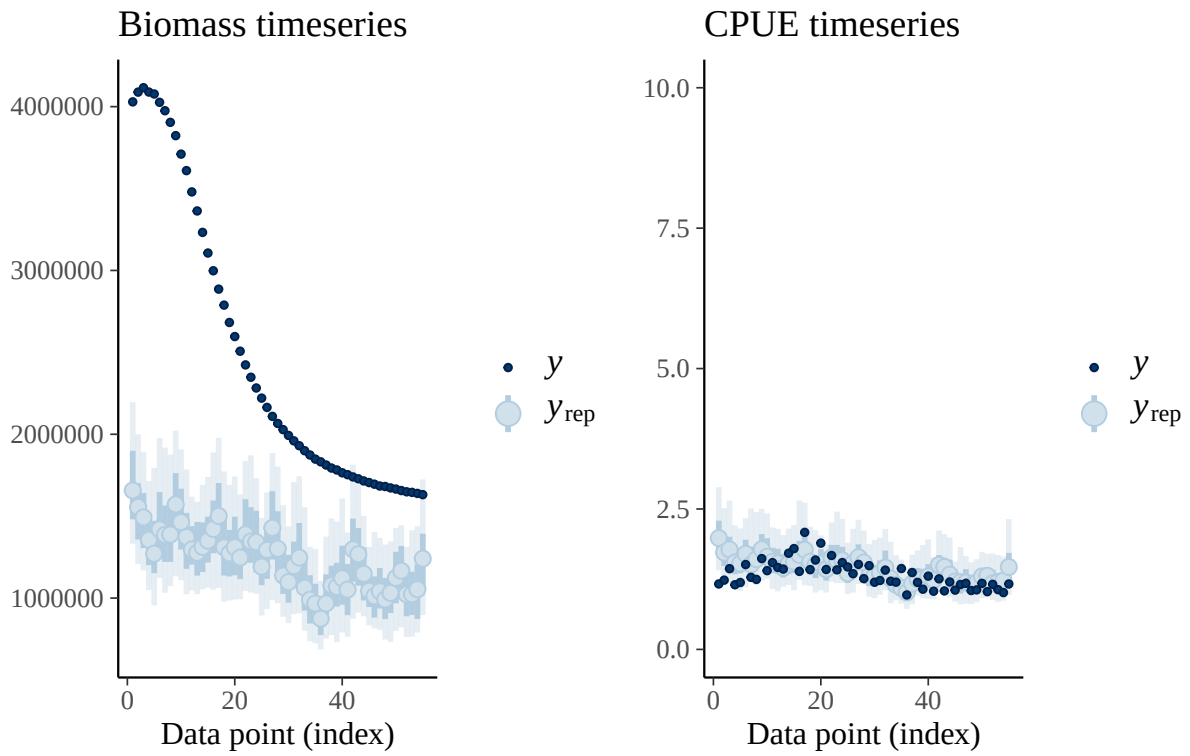


Constant

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
r	0.6757952	0.6651592	0.1203443	0.1155108	0.4990135	0.8851503	1.006069	999.7330	1420.0536
K	2665692.4470000	2629733.1500000	460411.5286268	457693.8166500	1986479.2100000	3470716.7150000	1.007394	615.6313	1256.3400
sigma_proc	0.1605307	0.1548605	0.0404587	0.0385593	0.1041564	0.2349537	1.005425	499.4871	1088.4718
sigma_obs	0.1796971	0.1792853	0.0308702	0.0294630	0.1283786	0.2305896	1.009422	506.7882	559.1998

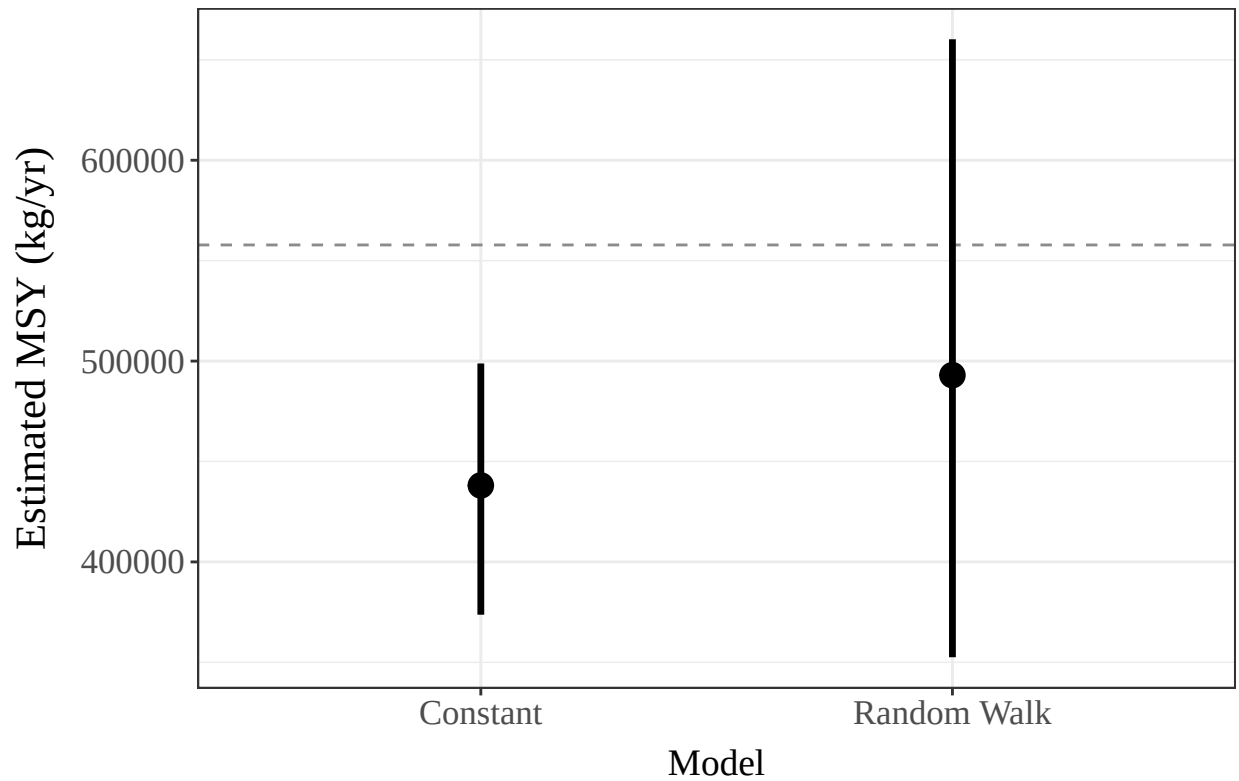


PPC: Constant q

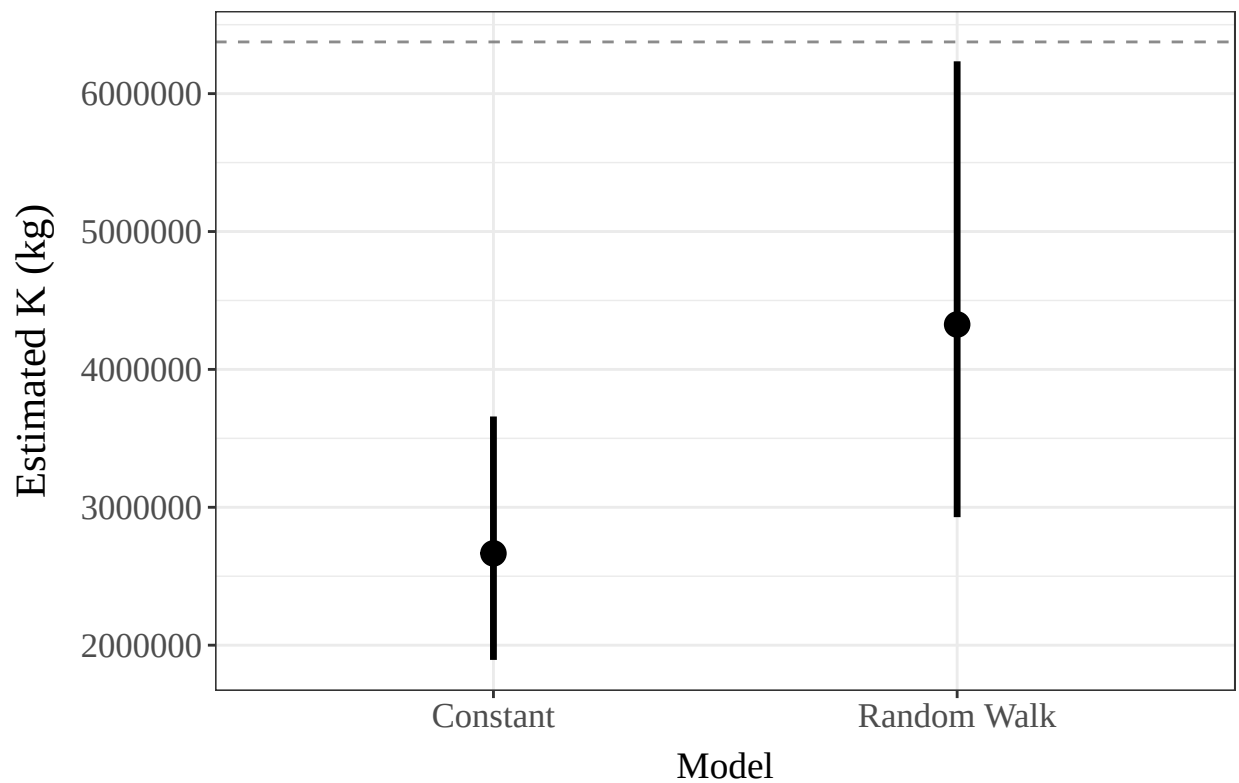


Comparison

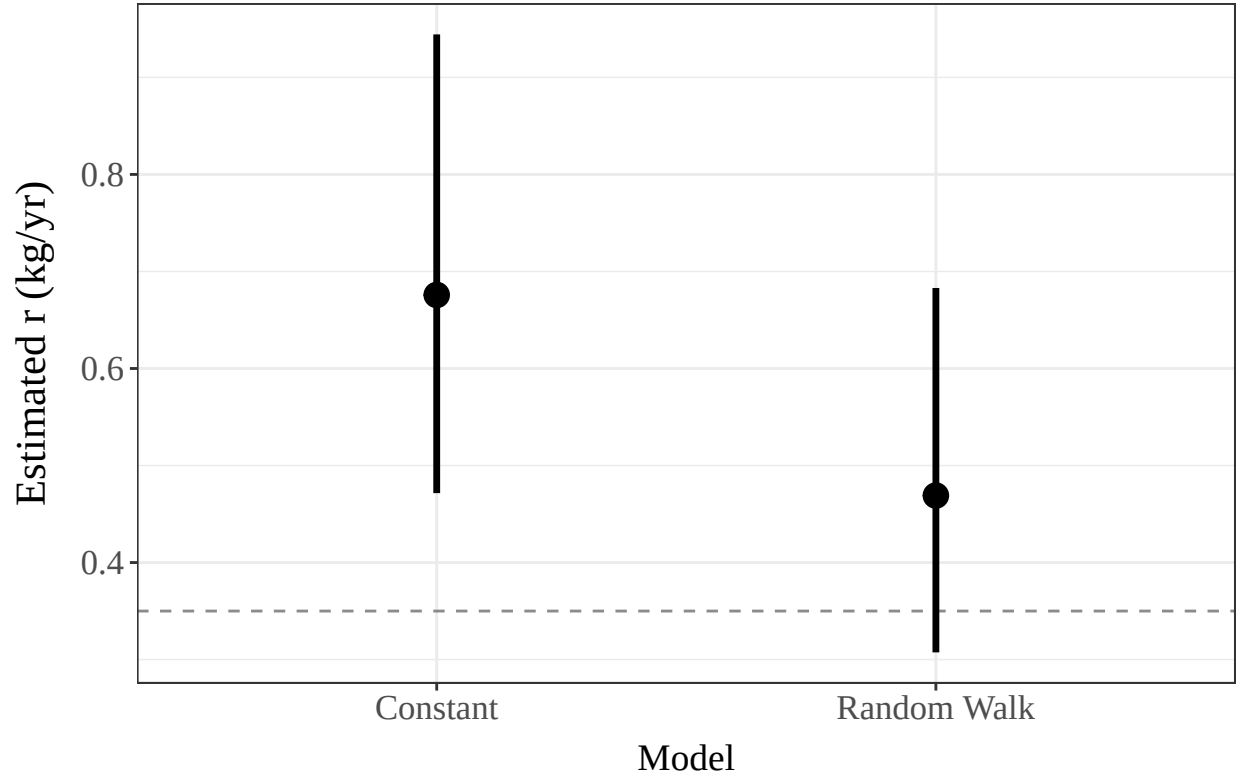
Estimated MSY compared to True MSY – Informative



Estimated K compared to True K



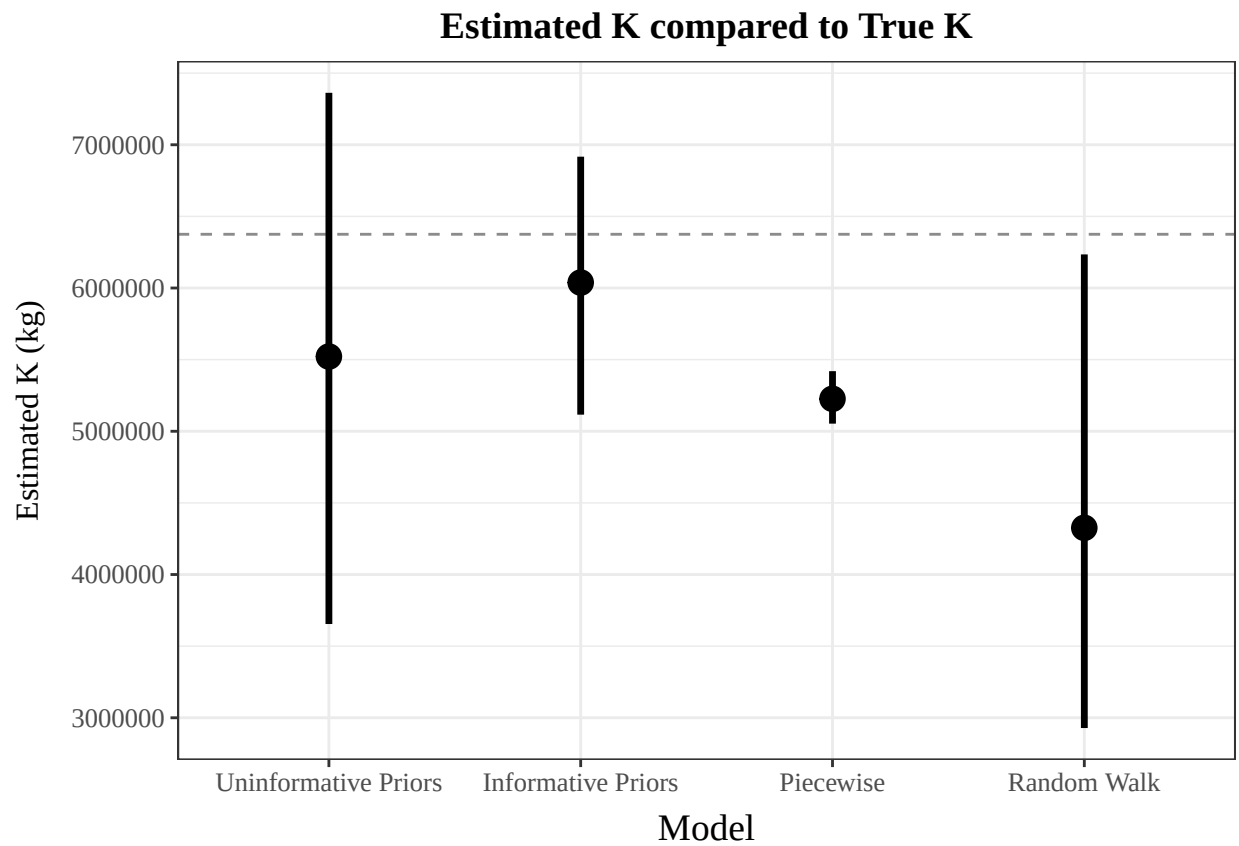
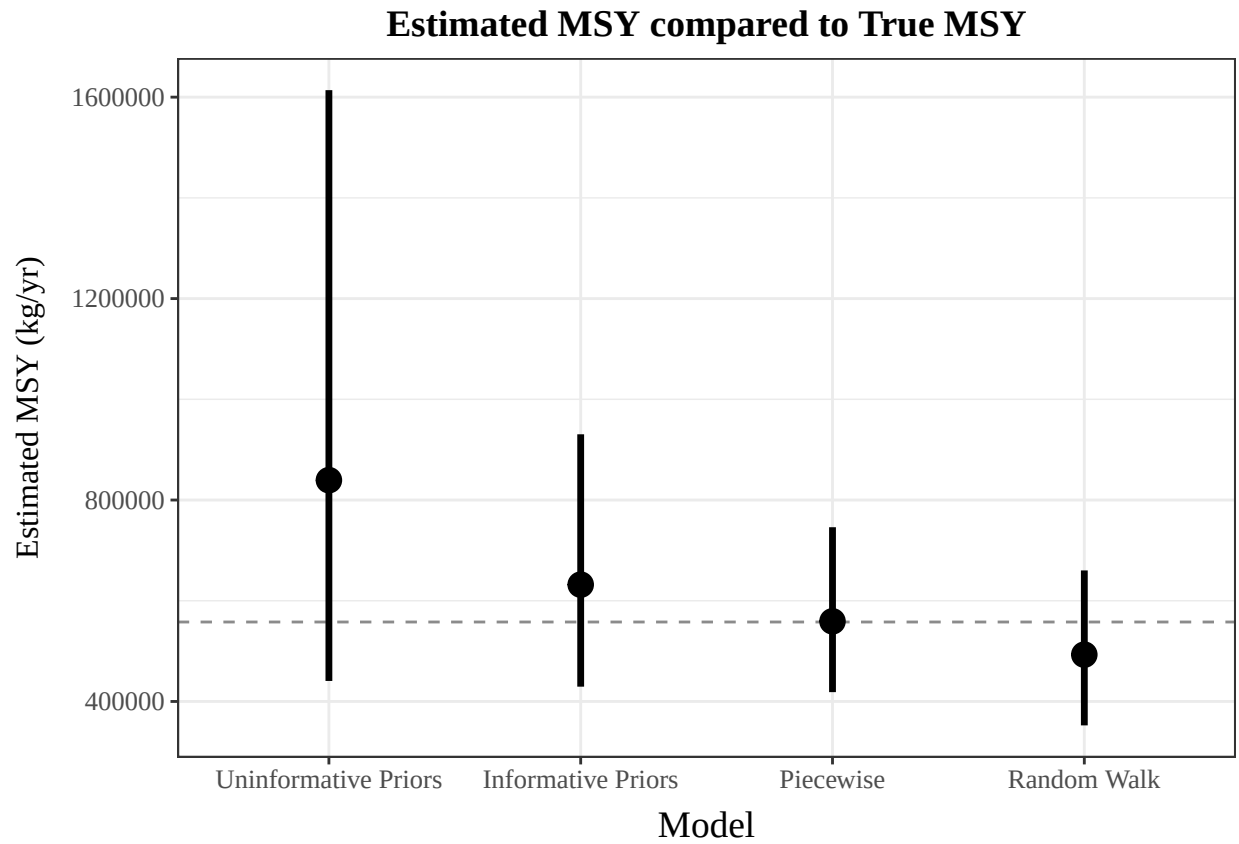
Estimated r compared to True r

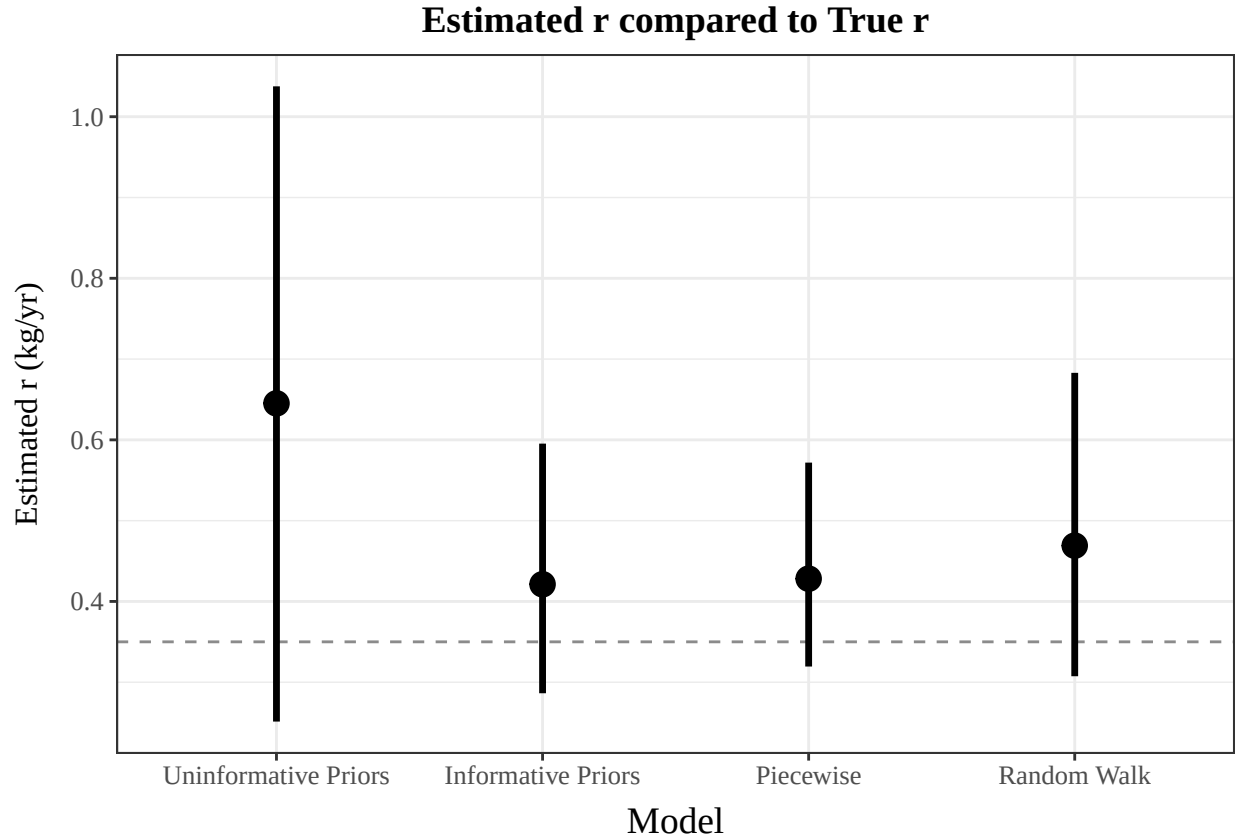


Model	WAIC	LOOIC
Constant q	25.27504	32.84268
Random walk q	258.27540	260.95066

Comparing the random walk and constant q Stan models using WAIC and LOOIC, the random walk model is preferred. This indicates that the random walk on q improves the predictive performance of a model when fitting to data that has adaptive fisher behavior.

Overall Model Evaluation





The Random Walk model and informative priors models appear to do the best at estimating MSY, accounting for K and r . While the piecewise model estimates MSY somewhat successfully it is severely biased in its estimate for K .

Discussion

Overarchingly, these results indicate that data that has a variable (rather than constant) q increases uncertainty when using standard methods for fisheries stock assessment. These have the potential to indicate that the stock is in better condition than it truly is and overestimate MSY, both of which may influence policy setting and result in overharvest.

Given that humans have learning capabilities and are able to identify areas in which fishing yields greater harvest for a given effort, it is likely that assuming a constant q in fisheries stock assessments is incorrect. While it presents a simpler mathematical model, which is beneficial for easily producing stock assessments, presents not only a concern of flattening the diverse skill sets and history of use of fishers, but additionally risks setting incorrect harvest targets which poses economic and ecological risks.

The use of informative priors, that suggest a strong biological understanding of the fisheries system, work to combat this and decrease the uncertainty in posterior estimates of stock parameters. However, the extent of remaining uncertainty emphasizes that improved biological knowledge alone cannot substitute for the incorporation of social fisher behavior. Putting a random walk on q performs the best in its ability to extract accurate estimates of MSY, but can be more time intensive to produce. Weighing the input costs of model development with the potential risks associated with less-accurate models is necessary to consider. Additionally, this highlights the limitations of packages like JABBA that improve the ease of the modeling process, but may lack ability to add increased nuance.

These show that it is necessary to consider both the behavior of fishers as well as biological parameters when conducting stock assessments. Given that fisheries policy utilizes levers (TAC, ITQs, etc.) that aim to target human behavior, an understanding of human behavior as well as fisheries parameters allows for decreased uncertainty and greater accuracy in resultant stock assessments.

Future steps may involve translating the posterior MSY estimates to policy parameters. These parameters can be implemented back into POSEIDON, enabling greater understanding of the ramifications associated with uncertain estimates, both ecologically and economically.